

JACK PEDAL

ESP32-S3 + ADS1220 | 0-10 bar pressure sensor

Required sensor power: 3.3 V DC

Sensor signal output: 0.4-2.4 V DC = 0-10 bar

Disconnect power before wiring. No resistor dividers are needed.

1 SENSOR CONNECTIONS

Sensor terminal		Connect to
Power (+): 3.3 V		ESP32 3V3
Ground (-)		Common GND
OUT: 0.4-2.4 V		ADS1220 AIN1

Use the sensor label to identify wires; do not assume wire colors.

2 ADC POWER AND SPI CONNECTIONS

ESP32-S3		ADS1220
3V3		AVDD, DVDD, REFP0
GND		AGND, DGND, REFN0
GND		CLK (internal ADC clock)
GPIO10		CS
GPIO9		DRDY
GPIO13		SCLK
GPIO12		MISO / DOUT
GPIO11		MOSI / DIN

All grounds must connect together. Power the sensor and ADC at 3.3 V.

Remove old divider resistors on AIN1, MISO and DRDY; connect directly.

3 QUICK VOLTAGE CHECK

With power on, measure AIN1 relative to GND:

0 bar	5 bar	10 bar
0.4 V	1.4 V	2.4 V

USB AND SOFTWARE

UART/CP2102 USB: upload firmware and connect the Windows app.

Native USB: game controller. Close Serial Monitor before using the app.

Use the 10 bar firmware; capture MIN, then MAX. 10 bar = about 145 PSI.